



Obesity Treatment: Lifestyle, Behavioral, and Medications

Note: This reference document contains the content of the Canvas page for “Obesity Treatment: Lifestyle, Behavioral, and Medications” with minor modifications for presentation in a document format. Refer to this session’s Canvas page for the full interactive experience or if any hyperlinks do not function properly.

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Overview

Obesity is a complex disease with a multifactorial etiology that has significant implications to healthcare systems. As such, understanding the multifaceted nature of obesity is crucial for effective treatment and management. In this workshop, students will be introduced to an integrated approach managing obesity, beginning with an exploration of its epidemiology, including both modifiable and non-modifiable risk factors, then moving into the comprehensive overview of obesity treatment, focusing on common pharmacological strategies and bariatric surgery and its role in managing severe obesity. Students will understand the potential biochemical complications of these treatments and be introduced to nutritional guidelines to mitigate their risks.

Community Medicine Learning Objectives

By the end of this workshop students should be able to:

1. Describe obesity as a complex, chronic disease.
2. Illustrate an understanding of the epidemiology of Obesity including modifiable and non-modifiable risk factors.
3. Describe obesity classification and staging systems including the Kings, Edmonton and American Obesity Society staging systems.
4. Assess a patient's readiness to change using the 5As model for behavior change.

Biochemistry Learning Objectives

By the end of this workshop students should be able to:

1. Describe the purpose of bariatric surgery as part of a larger strategy of weight control for severely obese patients.
2. Describe the metabolic consequences of bariatric surgery, particularly in diabetic patients.
3. Describe the pre-surgical diet and dietary requirements for a bariatric surgery patient.
4. Describe post-surgical diet, dietary requirements, and exercise recommendations for a bariatric surgery patient, including carbohydrate, protein, fat, vitamin, and liquid components.
5. Describe symptoms of food intolerance including vomiting and Dumping Syndrome, including causes, signs, symptoms, and prevention.

Pharmacology Learning Objectives

By the end of this workshop students should be able to:

1. List the major classes of drugs used to treat obesity.
2. Explain the mechanisms of action of specific anti-obesity medications.
3. Explain the adverse effects of specific anti-obesity medications
4. Explain the pharmacokinetics of specific anti-obesity medications



Clinical Skills Learning Objectives

By the end of this workshop students should be able to:

1. Define metabolic syndrome.
2. Explain the clinical presentations of metabolic syndrome.
3. Explain the pathophysiology of metabolic syndrome.
4. Describe the diagnosis of metabolic syndrome.

Preparation Content

For this session, content has been sectioned into separate sections. Visit these sections below to review content for each topic.

- Fusion Session | Workshop: Obesity Treatment: Lifestyle, Behavioral and Medications Part 1 of 4: Community Medicine
- Fusion Session | Workshop: Obesity Treatment: Lifestyle, Behavioral and Medications Part 2 of 4: Weight Management and Bariatric Surgery
- Fusion Session | Workshop: Obesity Treatment: Lifestyle, Behavioral and Medications Part 3 of 4: Pharmacotherapy
- Fusion Session | Workshop: Obesity Treatment: Lifestyle, Behavioral and Medications Part 4 of 4: Clinical Skills

Obesity Treatment: Lifestyle, Behavioral, and Medications Part 1 of 4: Community Medicine

Overview

Learning Objectives

This section covers the following learning objectives:

1. Describe obesity as a complex, chronic disease.
2. Illustrate an understanding of the epidemiology of obesity including modifiable and non-modifiable risk factors.
3. Describe obesity classification and staging systems including the Kings, Edmonton and American Obesity Society staging systems.
4. Assess a patient's readiness to change using the 5As model for behavior change.

Resources

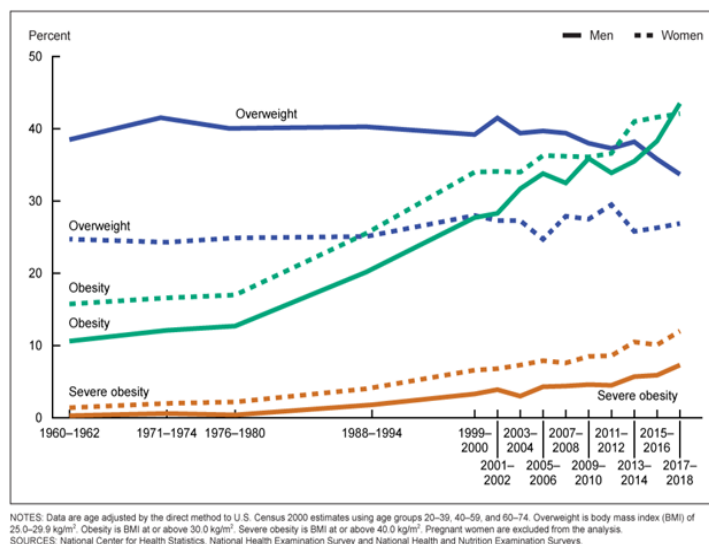
Required

Sheer AJ, Lo MC. [Counseling Patients With Obesity](#). [Updated 2023 Feb 9]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan.

Obesity: A Complex, Chronic Disease

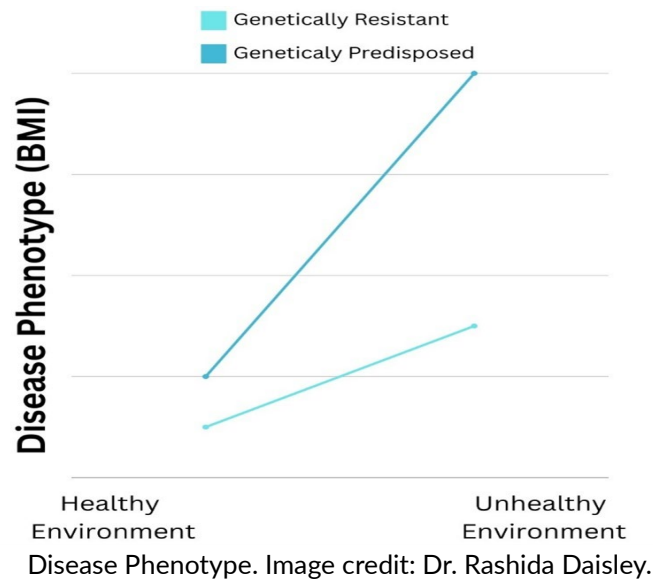
According to the data collected from the 2017-2018 National Health and Nutrition Survey (NHANES), an estimated 41.5% of US adults are affected by obesity, with 9% of those being affected by severe obesity. The graph below shows the trends in overweight/obese prevalence in the United States from 1960 to 2018. From this graph we can see the prevalence of these conditions has exponentially increased in recent years with women being more significantly affected when compared to men.

Environmental, behavioral, and genetic factors have been shown to contribute to the development of obesity. By 2025, global prevalence of obesity in adults is projected to reach 18/5 in men and 21% in women.



Age-adjusted trends in overweight, obesity, and severe obesity among men and women aged 20-74: United States, 1960-1962 through 2017-2018. Image credit: CDC.

Obesity is not only a risk factor for other diseases, but also a disease in its own right. Obesity was added to the sixth International Classification of Diseases in 1948, but only in 2013 was it recognized as a disease by the American Medical Association. As illustrated by the statistics above, Obesity is now one of the 21st century's greatest public health challenges, rising fastest in low- and middle-income countries, and requires urgent attention not only at the public health level but also at the level of the patient. The disease of obesity is a chronic progressive disorder of the energy homeostasis system, resulting in an individual having an excess of adiposity or body fat to the extent that health may be impaired. Obesity is a result of multiple environmental and genetic factors. The relative contributions of genetics and environment to the etiology of obesity have been evaluated in many studies. In a given population, some people are genetically predisposed to develop obesity, but that genotype may be expressed only under certain adverse environmental conditions, such as high-fat diets and sedentary lifestyles. Individuals genetically predisposed to a multifactorial disease, i.e., obesity, will likely manifest the disease phenotype in a healthy environment. In an adverse, i.e., obesogenic, environment, the genetic susceptibility will be expressed as the disease phenotype, but not in the genetically resistant population.



Obesity: A Complex, Chronic Disease

Complications of Obesity

Obesity is associated with an increased risk of all-cause mortality, with CVD and malignancy being the most common causes of death. The excess adiposity that characterizes obesity can cause complications through anatomical and metabolic effects.

The table below summarizes the prevalence of various complications in patients with obesity.

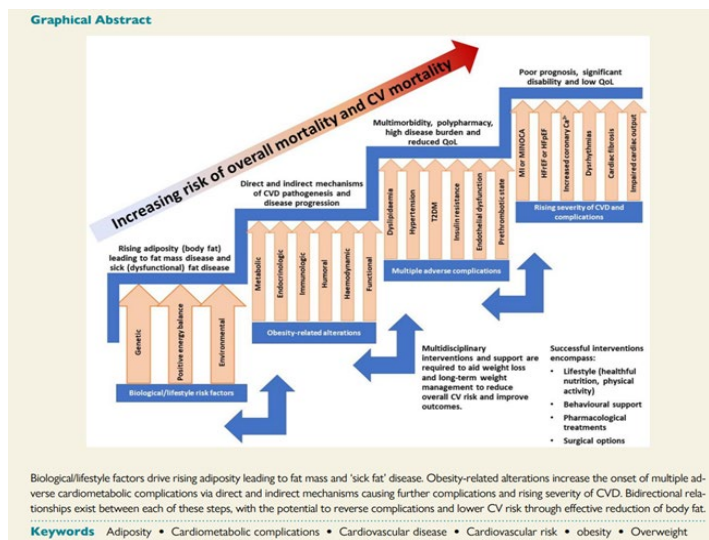
Complication	Prevalence in Patients living with Obesity
Major Depression	19%
Non-Alcoholic Fatty Liver Disease	19%
Ischemic Heart Disease 8%	8%
Polycystic Ovarian Syndrome	9%
Stroke	3%



Complication	Prevalence in Patients living with Obesity
Type 2 Diabetes Mellitus	21%
Deep Vein Thrombosis	2%
Cancer	4.7%-9.6%
GERD	35%
Obstructive Sleep Apnea	40%
Hypertension	51%
Knee Osteoarthritis	52%

Obesity and Cardiovascular Disease

Cardiovascular disease (CVD), including myocardial infarction (MI), stroke, and heart failure (HF), is a leading cause of death and disability in people living with overweight or obesity and early detection of cardiovascular (CV) risk is critical in reducing mortality and preserving quality of life (QoL). Accumulation of excess body fat (adiposity), particularly central/abdominal fat, is a well-established health risk. While body mass index (BMI), is used most widely as a pragmatic indicator of excess weight, waist circumference (WC) and waist-hip ratio (WHR) are better predictors of fat distribution and central obesity and have a closer association with morbidity.



Source: Lopez-Jimenez, F., Almahmeed, W., Bays, H., Cuevas, A., Di Angelantonio, E., le Roux, C. W., Sattar, N., Sun, M. C., Wittert, G., Pinto, F. J., & Wilding, J. P. H. (2022). [Obesity and cardiovascular disease: mechanistic insights and management strategies](#). A joint position paper by the World Heart Federation and World Obesity Federation. *European journal of preventive cardiology*, 29(17), 2218–2237.

Multiple obesity-related mechanisms drive structural, functional, humoral, and hemodynamic alterations believed to underpin the development of CVD, most notably coronary artery disease (CAD), HF, and arrhythmias in patients with obesity. There are two (2) broad ways in which obesity can influence the development of cardiovascular disease:

Deranged Endocrine & Immune Responses

Sick Fat Disease (SFD) (Adiposopathy)

Adiposopathy ("sick fat") is defined as adipocyte/adipose tissue dysfunction caused by positive caloric balance and sedentary lifestyle in genetically and environmentally susceptible individuals:

- Endocrine/metabolic:
 - Elevated blood glucose
 - Elevated blood pressure
 - Dyslipidemia
 - Other metabolic diseases

Abnormal and Pathologic Physical Forces

Fat Mass Disease (FMD)

Fat mass disease includes the biomechanical complications of obesity, such as obstructive sleep apnea (OSA) and joint stress and damage:



- Biomechanical/structural:
 - Stress on weight-bearing joints, Immobility
 - Tissue compression (i.e., sleep apnea, gastrointestinal reflux, high blood pressure, etc.)
 - Tissue friction (i.e., intertrigo, etc.)

Counseling Patients Living with Obesity

Weight stigma is a significant barrier to healthcare access for persons living with obesity. Therefore, it is critical that health professionals engage with the population with care, empathy and understanding. This begins with the infrastructure of clinical settings; clinicians can aim to ensure their practices are “obesity friendly” in the following ways:

1. Sturdy armless chairs, wide chairs with arms, and/or firm sofas in waiting rooms and exam rooms.
2. Sturdy wide exam tables.
3. Sturdy stools or steps with handles to help patients climb onto the exam table.
4. Extra-large patient gowns
5. Split toilet seats; provide a specimen collector with a handle.
6. Ensuring the facility is equipped with the following:
 - Large adult blood pressure cuffs or thigh cuffs that will fit on patients with an upper-arm circumference greater than 34 cm.
 - Large vaginal specula
 - Weight scales with the capacity to measure patients who weigh more than 400 pounds and ensuring that weight scales are located in a private area and not in the waiting room.

Using Obesity-Friendly Language

“People-first” language recognizes the preference of not identifying or labeling individuals by their disease. Thus, “patients with overweight or obesity” or “patients with pre-obesity or obesity” are preferred over “obese patient”.

Terms discouraged and best avoided include:

- Morbidly obese
- Obese
- Fat
- Heavy
- Large size

5 A'S Model for Discussing Obesity with Patients



Weight is a sensitive issue, and so conversations about weight must be sensitive and non-judgmental. The 5As model is a framework that can be used to guide the sensitive discussion.

- Watch this video for more information: [The 5As of Obesity Management](#) (YouTube, 00:05:00)
- Visit this page to download the Practitioner's Guide for Personal Use. Study how it can be used during discussions with persons living with obesity: [Practitioner Guide](#).

Obesity Classification and Staging Systems

In Cardio Module in Semester 3, we discussed body mass index (BMI) classification, and you were introduced to the concept of staging of obesity.

Body Mass Index (BMI) is the most widely used method of classifying obesity. BMI is a measure of size not health and so has some limits as a diagnostic tool. For example, the commonly utilized analogy of the BMI of athletes such as body builders may be misclassified due their high muscle mass. Although other useful anthropometric measures such as waist circumference, waist to hip ratio and body fat percentage can be useful at indicating distribution of fat, they do not provide any information on presence of functional limitations, comorbidities, or complications of obesity. Staging systems such as the King's Edmonton's and AACE systems provide several obesity staging systems have been developed to assist clinicians in characterizing obesity severity and aid in clinical decision-making. There are several staging systems that exist.

- Download the full text article: [Disease Severity and Staging of Obesity: a Rational Approach to Patient Selection](#).
 - Study the **King's Obesity Staging Criteria**: Table 1 and Figure 6.
 - Study the **Edmonton's Obesity Staging System**: Figure 7.
- Download this article: [AACE What Is the Disease of Obesity? Clinical Evaluation](#)
 - Study the **weight-related complications** relevant to the AACE guidelines: Slides 13-18.



Obesity Treatment: Lifestyle, Behavioral, and Medications Part 2 of 4: Weight Management and Bariatric Surgery

Overview

This portion of content includes an overview of the purpose and goals of bariatric surgery, followed by an exploration of its impact on weight management and metabolism. Next, pre-surgery nutrition is addressed, including guidelines for the weeks leading up to surgery and specific dietary requirements the day before and up to the procedure.

Following surgery, the focus shifts to post-surgical diets, with attention given to different procedures like laparoscopic band and gastric bypass. The progression through the post-surgery phases, from phases 1 to 5, is discussed, along with the importance of vitamin supplementation during recovery.

The final sections cover the necessary post-surgery lifestyle changes, types of foods to prioritize, and potential dietary complications, such as vomiting and Dumping Syndrome.

Learning Objectives

This section covers the following learning objectives:

1. Describe the purpose of bariatric surgery as part of a larger strategy of weight control for severely obese patients.
2. Describe the metabolic consequences of bariatric surgery, particularly in diabetic patients.
3. Describe the pre-surgical diet and dietary requirements for a bariatric surgery patient.
4. Describe post-surgical diet, dietary requirements, and exercise recommendations for a bariatric surgery patient, including carbohydrate, protein, fat, vitamin, and liquid components.
5. Describe symptoms of food intolerance including vomiting and Dumping Syndrome, including causes, signs, symptoms, and prevention.

Resources

Required

Videos

- [Gastric Bypass](#) (YouTube, 00:02:32)
- [Gastric Sleeve](#) (YouTube, 00:01:24)
- [Gastric Band/Lap Band](#) (YouTube, 00:01:31)

Readings



- Sherf Dagan, S., Goldenshluger, A., Globus, I., Schweiger, C., Kessler, Y., Kowen Sandbank, G., Ben-Porat, T., & Sinai, T. (2017). [Nutritional Recommendations for Adult Bariatric Surgery Patients: Clinical Practice](#). *Advances in nutrition* (Bethesda, Md.), 8(2), 382–394.
- Cleveland Clinic medical. (2024, June 4). [Bariatric \(weight loss\) surgery: Types & requirements](#). Cleveland Clinic.
- [Gastric Dumping Syndrome](#)

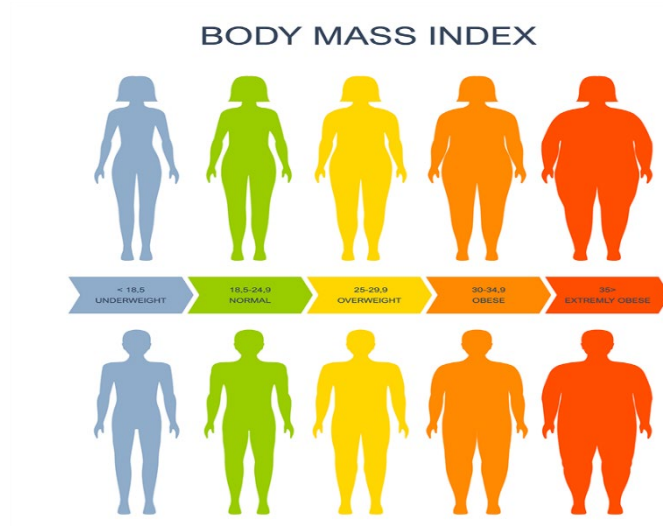
Goals of Weight Management Surgery

Objective 1

Depending upon the culture in which a patient lives, obesity often results in other psychosocial problems for the obese person. The burden of obesity is further magnified by both the failure of a given populace to recognize a genetic predisposition to obesity, and by the daily struggles of many obese people to succeed against the effects of an “obesogenic environment”.

Obesity is a complex, multifactorial, and chronic disorder that often leads to serious psychosocial and physical health consequences. Depending on the cultural context in which an individual lives, obesity can result in additional psychological and social challenges, further complicating the individual's condition. The burden of obesity is compounded by society's frequent failure to acknowledge the genetic predispositions that contribute to the condition and by the daily challenges obese individuals face in navigating an "obesogenic environment." This environment fosters conditions that make maintaining a healthy weight difficult, leading to an increased prevalence of obesity-related health issues, such as hypertension, diabetes, strokes, and heart disease.

For adults, the primary goal of weight control is often the prevention of obesity, which in turn can reduce the severity of these associated health conditions. Obesity is commonly defined using Body Mass Index (BMI), and maintaining a BMI within the normal range is essential for minimizing the risk of related pathologies. When obesity is already present, various weight management strategies, including lifestyle and dietary changes, can help individuals return to a healthier BMI, thereby improving their quality of life. For some, bariatric surgery may be a viable option, but it is just one tool in a comprehensive approach to managing obesity. Addressing obesity requires a multifaceted strategy that combines prevention, lifestyle modifications, and, when necessary, medical interventions to effectively manage and treat this serious health condition.



Body Mass Index. Image Credit: Adobe, used with permission.

Impact of Surgery on Weight

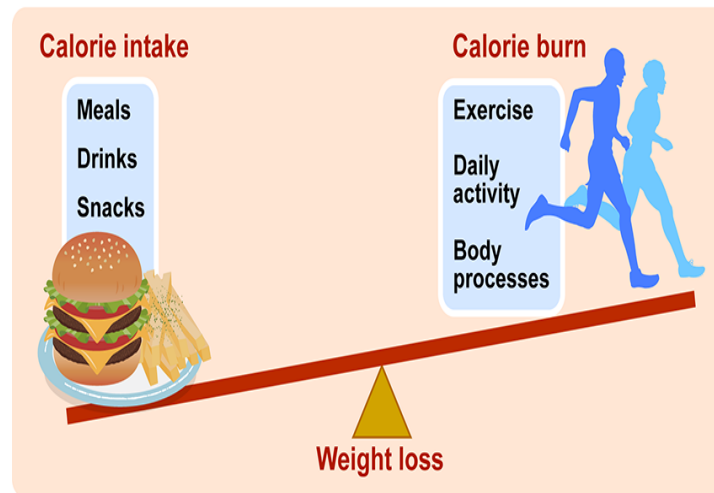
Objective 2

Though the primary focus of this presentation is on the dietary management of a patient before and after bariatric surgery, it is important to remember that there are also metabolic effects of this surgery. There is some debate in the medical community of a chicken-and-the-egg nature, as to the mechanism of this metabolic change. Is it all from changes in the responsiveness of GI-based factors such as Glucagon-like-peptide 1 (GLP-1)? Or is it all about the food restriction leading to these effects? While the debate continues, it is undisputed that there are specific benefits.

Bariatric surgery aims to reduce weight and then maintain weight loss through two related mechanisms:

1. Reduction of Food Intake
 - If food intake is less than energy expenditure, the patient will lose weight.

Weight Loss



Balance of calorie intake and burn. Image Credit: Aadtalem.

2. Modification of Metabolism through, for example, changes in insulin responsiveness and glucose tolerance, particularly in patients with type 2 diabetes mellitus.
 - Research has shown that there are metabolic benefits after bariatric surgery associated with the effect of incretins (e.g., GLP-1) and possibly other hormonal or neural changes.
 - For example, rapid and sustained improvements in glycemic control can be achieved within days of bypass surgery, before much weight loss. (These improvements might increase to levels seen in nondiabetic controls by 8 weeks post-surgery.)
 - Modest weight loss in people with type 2 diabetes who are overweight or obese is associated with improvement in glycemic control and better metabolic control of comorbidities like hypertension, obstructive sleep apnea, and dyslipidemia.
 - For more details of the different theories on how this works, see this article: Filip K. Knop, Roy Taylor; [Mechanism of Metabolic Advantages After Bariatric Surgery: It's all gastrointestinal factors versus it's all food restriction](#). Diabetes Care 1 August 2013; 36 (Supplement_2): S287–S291.

Pre-Surgery Selection Criteria

Criteria for Bariatric Surgery

- Obese (BMI > 40), or
- Obese (BMI > 35) and one or more comorbidities:
 - Heart disease



- Type 2 Diabetes
 - High blood pressure
 - Sleep apnea
 - Dyslipidemia
 - Long-standing obesity
- Weight < 450 lbs
- Obese (BMI > 30; Lap-band only after failure of diet and exercise).

Other Considerations

- Patient history of weight gain
- Patient motivation
- Access to social system
- Tobacco use
- Other substance abuse
- Psychopathologies
- Management of other medical conditions
- Lifestyle and employment stressors

Pre-Surgery Dietary Preparation

Part 1: Weeks Prior to Surgery

- Stop smoking 30 days prior
 - Helps with anesthesia, pain management
 - Helps with post-operative exercise regimen
- Reduce or eliminate caffeinated beverages
 - Drink plenty of water
- Calorie-controlled diet
 - Take multivitamins
 - Reduce liver size (less fat/glycogen)
 - Start good eating habits that will be needed post-surgery



Dietary preparation before weight loss surgery. Image Credit: Aadtalem.

Part 2 (most patients, but not all)

- Liquid, bowel-preparatory diet
- Greatly reduces the chance of contamination from stools in the bowel as well as the production of potentially explosive intestinal gases.
- Allows the bowel to rest after the surgery.
- Decrease high-residue foods or avoid them altogether.
- Suggested foods include broths, herbal teas, and sugar-free popsicles.

Note

There is no clear consensus that a liquid diet prior to surgery is necessary. For some surgeries such as laparoscopic band placement it is less important than for others that require working with the lower bowel. Clear liquids allowed include water, coffee, tea, clear broth or bouillon, Gatorade, PowerAde, carbonated and non-carbonated soft drinks, Kool-Aid or other flavored drinks, clear fruit juices such as apple or white grape juice, Jell-O, or popsicles.

Part 3: Preparation for Surgery

- No food at all on the day of surgery (starting midnight of the day before)
- Check with the doctor for exact requirements
- Includes none of the following:
 - Water
 - Gum
 - Candy



- Any other food
- Place patient on IV fluids prior to surgery



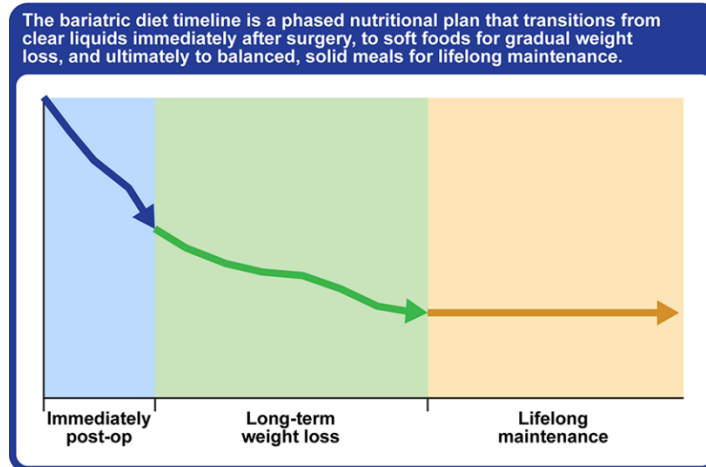
Weight Loss Surgery. Image Credit: Adobe, used with permission.

General Post-Surgery Dietary Preparation

General Dietary Considerations:

- The overall idea in post-bariatric surgery patients is to allow a gradual reintroduction to normal foods, with some permanent restrictions (e.g., hard seeds, nuts).
- Patients must learn a new pattern of food and liquid intake to prevent clinical conditions, which they likely will follow the rest of their lives.
- For those procedures which modify the small intestine very little, there is one diet plan which is commonly used.
- For others, the diet plan must accommodate for the missing functions of the intestinal tract.

Bariatric Diet Timeline



Bariatric Diet Timeline. Image Credit:

Post-Surgery General Dietary Changes

- Bariatric surgery impacts digestion and absorption of some nutrients more than others.
- Vitamins are impacted:
 - Give Multivitamins: 100-200% RDA
 - Of particular concern:
 - Thiamine, Vitamin D, Calcium, Vitamin B12, Folic Acid, and Iron
 - Thiamine loss can be exacerbated by chronic vomiting
 - Calcium citrate and Vitamin D to prevent bone disorders
 - With gastric bypass, additional B12, folate, and iron are necessary for red blood cell synthesis
- Also, in some surgeries, fat-soluble vitamins may be poorly absorbed
 - Supplement vitamins A, D, E, and K
- Macronutrients:
 - Protein:
 - Increase from 0.8 g/kg/day to 1-1.5 g/kg/day
 - Use $\frac{1}{4}$ to $\frac{1}{2}$ cup portion sizes per meal.
 - Use supplements, e.g., soy, egg white, and milk-based products
 - Protein isolates are lactose-free and well tolerated
 - Available in clear liquids, shakes, powders, and bars
 - (to prevent loss of lean body mass).
 - Liquids: Avoid dehydration – a key concern
 - Drink $\frac{1}{4}$ cup plain water every 15 minutes.
 - No sugar
 - No carbonation (causes expansion and stretching of the stomach making it quite uncomfortable)



- No caffeine (more likely to cause blood clots due to dehydration)
- Do not drink at meals – wait 30 mins to drink to avoid dumping syndrome.
- Can use sugar-free popsicles, broth, or decaffeinated tea or coffee

Foods to Avoid after Weight Loss Surgery



Foods to Avoid After Weight Loss Surgery. Image Credit: Adtalem.

Temporal Post-Surgery Diet Changes

Read the sections below to learn more about the phases of post-surgery diet changes.

Phase 1

- 0-6 hours post-op.
- Nothing.
- Maintain IV fluid administration as begun prior to surgery.

Phase 2

- 6 hours post-op to 2-3 weeks post-op
- Liquid Diet
 - e.g., Broth, unsweetened juice, milk, strained cream soup (2-3 ounces at a time), sugar-free gelatin
 - Liquid vitamin supplements
 - No caffeinated or carbonated drinks
 - These tend to stretch the stomach pouch needlessly
 - Avoiding distension of the stomach pouch prevents complications
- Allows for rest and recovery of the stomach and intestine



Phase 3

- 2–4 weeks post-op
- Required for intestinal tract to adjust to regular food again.
 - *Pureed foods*
 - e.g., Lean ground meats, beans, fish, egg whites, yogurts, diet soft drinks, vegetables, cottage cheese (continue small portions)
 - Liquid vitamins, rather than pills
 - Fat- and sugar-free are recommended
 - Be careful with spicy foods
 - Be cautious with carbonated beverages, due to possible distention

Phase 4

- 4–8 weeks post-op
- Soft solid foods
 - Add ground or finely diced meat, canned or soft vegetables and fruit, and cooked vegetables.
 - Chewable or liquid vitamins
- Water intake ~ 2L/day, no straws, a few sips at a time.
 - Drinking with a straw results in swallowing air, which causes distension.
- More intestinal adjustment, this time with the first addition of sugars in fruit.
 - Portion sizes need to stay very small (<4 oz).

Phase 5

- After 8 weeks post-op
- Regular food, with specific restrictions
 - Be careful about introducing or taking in high residue foods (high fiber and other non-digestible components, e.g., fruits, nuts, potato skins, figs)
 - Chewable vitamins, progress to tablets as tolerated
- Water intake continues at appx. 2L/day, no straws, a few sips at a time.
- Once adjustment to regular food is made, continue through life.
 - Small amounts of regular food, thus allowing the body to heal before adding high residue foods.
- Regular follow-up is important. This should include an assessment of:
 - Body fat loss
 - Potential anemia
 - Potassium and/or magnesium deficiencies
 - Folate and cobalamin levels
 - Iron deficiency



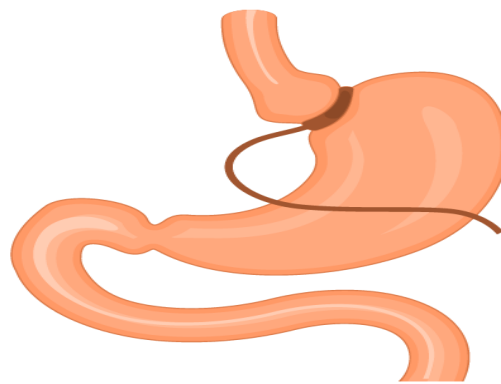
Exercise

- An exercise program is recommended in addition to dietary guidelines
- This serves to tighten the skin, improve muscle tone, lower blood pressure, and boost muscle strength
- It is important to build up exercise intensity gradually to a strenuous level
- Starting aerobic exercise even before the surgery is recommended

General Dietary Considerations by Procedure

Laparoscopic Band (Adjustable Gastric Banding)

- The fewest special dietary considerations, just maintain a strict eating plan with vitamin supplementation
- Diet can be adjusted if the band is ever removed

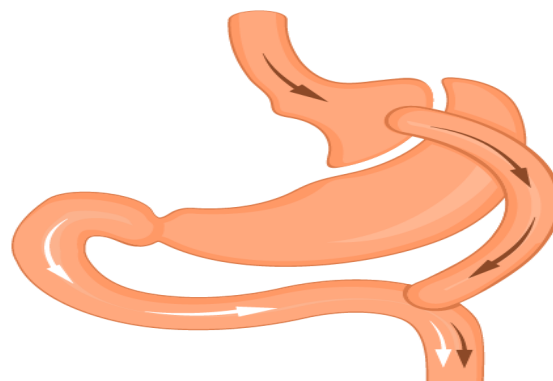


**Adjustable
Gastric Band (AGB)**

Laparoscopic Band (Adjustable Gastric Banding). Image Credit: Adobe, used with permission.

Gastric By-Pass (Roux-en-Y, proximal variant)

- Restrictive and possibly malabsorptive reduced intake
- Maintain strict eating plan, but dietary changes are for life
- Multivitamin and mineral supplementation required, including calcium, Vitamin D, and occasional injections of Vitamin B12 (cobalamin).
- Loss of intrinsic factor means that Vitamin B12 cannot be efficiently absorbed. Vitamin B12 is an unusual water-soluble vitamin as we have at least 1-3 months store in the liver, so it only needs to be administered once a month.



Roux-en-Y Gastric Bypass (RYGB)

Roux-en-Y Gastric Bypass. Image Credit: Adobe, used with permission.

Other Procedures

- There are other procedures which require more stringent post-op dietary management, e.g.,
 - duodenal switch with biliopancreatic diversion
 - sleeve gastrectomy
- For further information, refer to [Nutrition Guidelines for Sleeve Gastrectomy and Gastric Bypass](#). Brigham and Womens. (2017, June).

Post-Surgery Diet-Related Complications

Vomiting

- Vomiting is a cardinal sign of intolerance within the first few months of surgery.
- Often due to ingesting too large a meal.
- Ingestion of products that cause excess stretch (e.g., carbonated beverages) can also result in vomiting.
- Chronic vomiting can lead to thiamine deficiency, so thiamine supplementation is recommended.
- Recommended prevention protocol is ingestion of only small quantities of food per meal, and chew slowly.

Dumping Syndrome

- A clinical condition of malabsorption and/or osmotic overload in the small intestine,
 - Typically, due to too-fast or too-large meals, especially if high in sugar and fat



- This results in “dumping” the unprocessed food into the intestine too quickly.
- Intestinal/Colonic Bacteria metabolize unabsorbed nutrients, with overbloats and corresponding flatulence and cramping.

Signs and symptoms include nausea, diarrhea, tachycardia, sweating, intestinal gas production, and cramps.

Treatment and Prevention of Dumping Syndrome

- Take several small meals instead of any one large one.
- Do not drink fluids with meals.
- Eat and drink slowly, to prevent “dumping” of too much food into the intestine too quickly.
- Choose foods and liquids low in fat and sugar.
- Wait at least 30 minutes after meals before ingesting fluids.
- Take at least 30 minutes to eat meals.

Summary

- Bariatric surgery is a radical approach to weight loss in cases where diet and exercise have failed.
- Such surgeries require dramatic changes in diet and lifestyle.
- Glycemic control benefits can be seen shortly after surgery.
- Pre-surgery diets include omitting tobacco use and caffeine, increased water intake, reduced caloric intake, and multivitamins. This transitions to liquid diets and then the omission of food immediately pending surgery.
- Post-surgery diets move from liquid to semi-solid to solid foods. High-fiber and high-residue foods are initially omitted, as are foods high in fat and sugar.
- Eventually, after several months, the patient moves to regular foods as tolerated. Multivitamins should be given throughout.
 - Drinking water should be continuous during the day, but not at meals. Meals should be without liquids to avoid dumping syndrome.
 - Hard foods such as nuts, seeds, and popcorn are usually eliminated from the diet.
- Exercise is a key part of post-operative care, weight loss and glycemic control.
- Vomiting and dumping syndrome are two possible complications of surgery. Work with patients on how to avoid these.

Obesity Treatment: Lifestyle, Behavioral, and Medications Part 3 of 4: Pharmacotherapy

Overview

Several drugs have received approval from the US Food and Drug Administration to treat overweight or obesity conditions. It's crucial that these medications be combined with a regimen of a healthy diet, regular exercise, and behavior adjustments because relying on medication alone typically does not yield long-term effectiveness. When considering the start of pharmacotherapy for obesity, the potential advantages and disadvantages of weight loss drugs must be carefully weighed.

Learning Objectives

This section covers the following learning objectives:

1. List the major classes of drugs used to treat obesity.
2. Explain the mechanisms of action of specific anti-obesity medications.
3. Explain the adverse effects of specific anti-obesity medications
4. Explain the pharmacokinetics of specific anti-obesity medications

Goals of Therapy

Drug treatment for those who are overweight or obese should concentrate on achieving lasting weight loss and boosting general health. Read the sections below to learn more about the goals of therapy.

Reduce weight and maintain weight loss

The primary objectives of obesity drug therapy include not only reducing body weight but also ensuring that weight loss is sustained over time. This dual focus addresses the immediate health concerns associated with excess weight and combats the common trend of weight regain, which can undermine health benefits achieved through initial weight loss.

Improve health status

A key goal of obesity drug therapy is to enhance overall health. This includes mitigating obesity-related health risks such as diabetes, hypertension, and cardiovascular disease. Effective treatment aims not just at weight reduction but also at improving metabolic health, increasing physical fitness, and boosting mental well-being. By targeting these

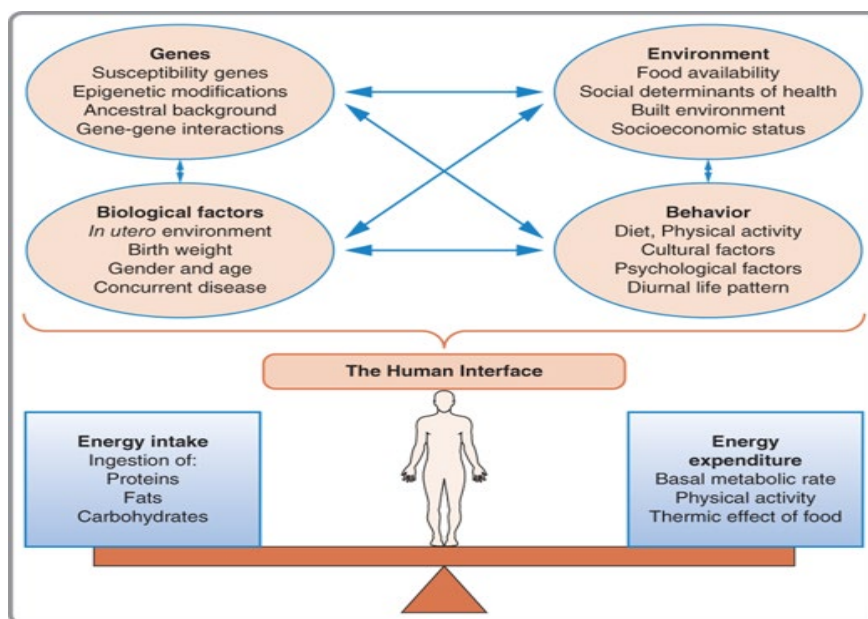
areas, obesity pharmacotherapy seeks to elevate the quality of life and decrease the likelihood of chronic disease.

Minimize adverse effects

An essential objective of obesity drug therapy is to minimize adverse effects associated with treatment. This entails selecting medications that not only effectively aid in weight loss but also have a favorable safety profile, ensuring that the benefits of weight reduction outweigh any potential risks. Careful monitoring and individualized treatment plans are vital to detect and manage any side effects early, enhancing patient comfort and adherence to the therapy.

Pathophysiology of Obesity

Obesity constitutes a disorder of energy equilibrium resulting in excessive adipose tissue accumulation, sufficient to detrimentally affect health. Analogous to other chronic conditions, the pathogenesis of obesity is multifaceted, encompassing interplays between genetic predispositions, biological determinants, environmental influences, and behavioral factors.



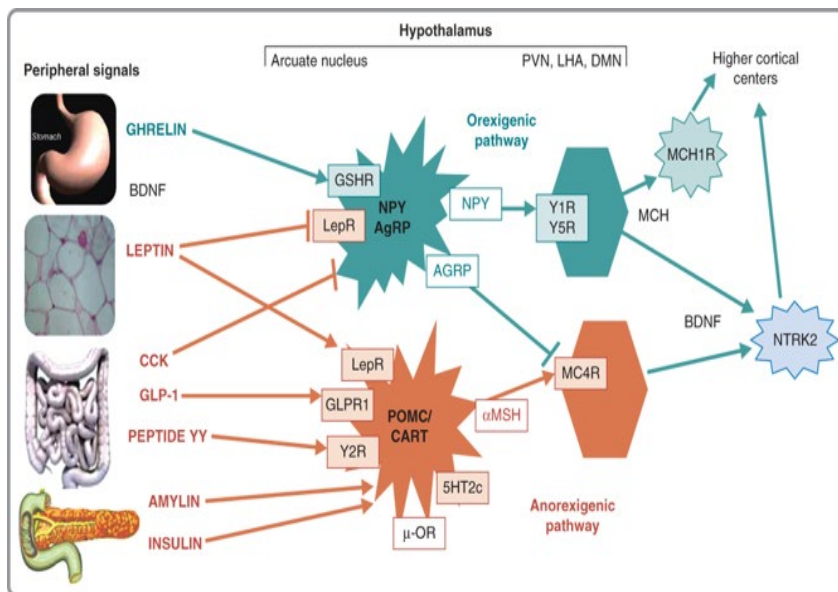
Source: V. Fuster, J. Narula, P. Vaishnava, M.B. Leon, D.J. Callans, J. Rumsfeld, A. Poppas: Fuster and Hurst's The Heart, 15e Copyright © McGraw Hill Education. All rights reserved.

Pathogenesis of Obesity. Image credit: McGraw Hill, used with permission.

Regulation of Caloric Intake by Satiety Hormones and Hypothalamic Feeding Centers

Satiety hormones are released from peripheral tissues and act on cells in the arcuate nucleus of the hypothalamus. The orexigenic hormone (ghrelin from the stomach) acts on

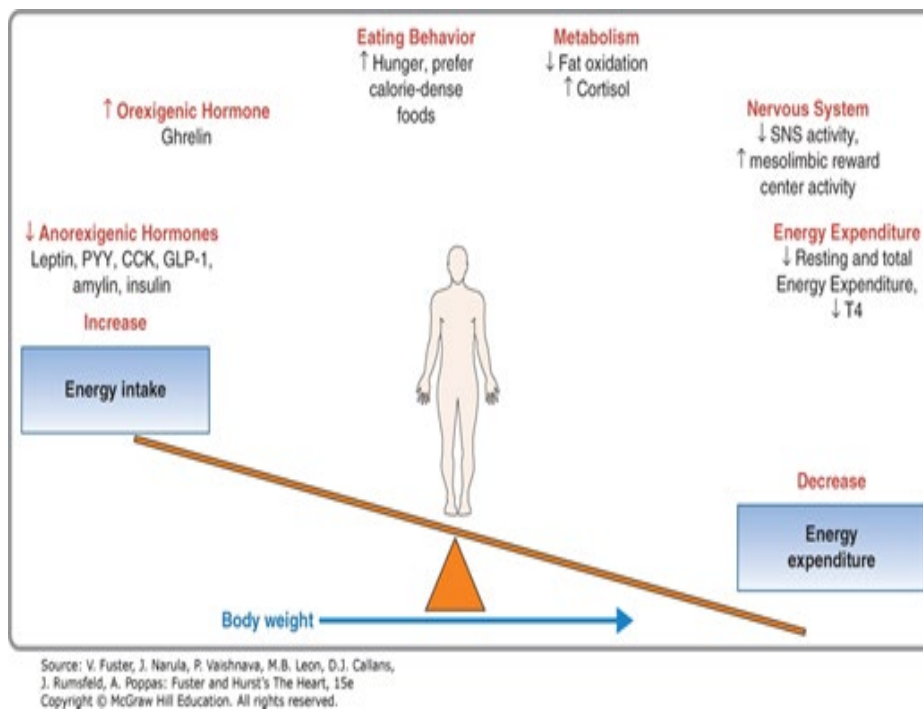
cells releasing NPY as a neurotransmitter. Anorexigenic hormones (leptin from fat and CCK, GLP-1, peptide YY, and amylin from the gastrointestinal tract) act on cells producing the neurotransmitter α MSH. Both cell types signal to higher cortical centers to regulate hunger and satiety. (CCK, cholecystokinin; GLP-1, glucagon-like peptide 1; NPY, neuropeptide Y; AgRP, agouti-related peptide; POMC, proopiomelanocortin; CART, cocaine and amphetamine-regulated transcript; α MSH, alpha melanocyte-stimulating hormone; PVN, paraventricular nucleus; LH, lateral hypothalamus; DMN, dorsomedial nucleus; GSHR, growth hormone secretagogue receptor.)



Regulation of caloric intake by satiety hormones and hypothalamic feeding centers. Image credit: McGraw Hill, used with permission.

Maladaptive Responses to Weight Loss in Patients with the Disease of Obesity

Following a weight-loss intervention, multiple pathophysiological processes act to increase energy intake and decrease energy expenditure, and to drive weight regain and return to the previous high level of body weight. Anorexigenic hormones decrease, and the orexigenic hormone ghrelin increases, resulting in greater hunger. Other mechanisms reduce the resting energy expenditure, further promoting a positive energy balance. PYY, peptide YY; CCK, cholecystokinin; GLP-1, glucagon-like peptide 1; SNS, sympathetic nervous system; T4, thyroxine.



Maladaptive Responses to Weight Loss in Patients with the Disease of Obesity. Image Credit: McGraw Hill, used with permission.

Classification of Drugs Used to Treat Obesity

First-Line Agents	Alternative Agents
Tirzepatide	Phentermine-topiramate
Semaglutide	Naltrexone-bupropion
Liraglutide	Orlistat

Read the sections below to learn more about each of these drugs.

Tirzepatide

Mechanism of Action

Tirzepatide is a glucose-dependent insulintropic polypeptide (GIP) receptor and glucagon-like peptide-1 (GLP-1) receptor agonist. Through the GLP-1 Receptor agonistic activity it promotes an anorexigenic effect, significantly attenuating appetite and reducing caloric intake. Activation of central GLP-1 receptors modulates neuroendocrine pathways involved in satiety signals, enhancing satiation and prolonging satiety, thereby facilitating hypophagic behavior. Through the GIP Receptor agonistic activity, it contributes to the modulation of energy expenditure. This effect, coupled with GLP-1 receptor-mediated actions, may synergistically enhance metabolic rate and fat oxidation.

Pharmacokinetics

Tirzepatide is administered subcutaneously via an injection. It is not available in oral form at this time. Tirzepatide has a bioavailability of approximately 80% and half-life of 5 days, allowing once-weekly dosing, and is cleared in urine and feces in the form of metabolites.

Adverse Effects

Based on available data, most users do not experience significant adverse drug reactions. The primary adverse effects reported are gastrointestinal in the form of decreased appetite, nausea, and diarrhea.

Semaglutide & Liraglutide

Mechanism of Action

Semaglutide and liraglutide are categorized as agonists of the glucagon-like peptide-1 (GLP-1) receptor. Within the ambit of adiposity management, their pharmacodynamic action principally involves the modulation of satiety and the diminution of caloric consumption, culminating in adipose tissue reduction. These pharmacotherapeutic agents exert their effects by engaging the hypothalamic centers and additional cerebral regions implicated in the homeostatic regulation of appetite and ingestive behavior. Through the agonistic simulation of endogenous GLP-1 activity, semaglutide and liraglutide potentiate satiogenic signals while attenuating orexigenic cues, thereby effectuating a decrement in energy intake. Moreover, they induce a retardation of gastric motility, augmenting postprandial satiation and contributing to sustained anorectic sensations.



Pharmacokinetics

Many formulations of GLP-1 agonists, all of which historically were injectable and administered subcutaneously due to poor oral bioavailability. Recently, the FDA approved an oral formulation of semaglutide.

Adverse Effects

The most frequently exhibited side effects from GLP-1 agonists include nausea, vomiting, and diarrhea that could lead to an acute kidney injury due to volume contraction. Dizziness, mild tachycardia, infections, headaches, and dyspepsia may also occur.

Phentermine-Topiramate

Mechanism of Action

Phentermine is a sympathomimetic amine anorectic. It has a similar mechanism of action as amphetamine in that it is an agonist at the trace amine associated (TAAR1) receptor site, stimulating the release of norepinephrine and epinephrine. It is a central nervous system stimulant.

Topiramate is an anticonvulsant lowering the seizure threshold, stabilizing membrane by acting on high-voltage-activated calcium channels and voltage-gated sodium channels and its augmenting effect on GABA-A receptors. Topiramate is also a weak carbonic anhydrase inhibitor, and it also antagonizes glutamate receptors. Topiramate augments appetite suppression and satiety enhancement based on the combination of the mechanisms above.

Pharmacokinetics

Phentermine has two metabolic pathways: hydroxylation and oxidation. Cytochrome P450 (CYP) 3A4 primarily metabolizes phentermine. Topiramate does not show extensive metabolism. Topiramate is primarily metabolized by hydroxylation, hydrolysis, and glucuronidation. Topiramate inhibits CYP2C19 enzymes and induces the CYP3A4 enzymes.

Adverse Effects

The most common adverse events reported in clinical trials are dry mouth, constipation, and paresthesia. Topiramate may cause severe hypohidrosis and hyperthermia. Avoid combination with anticholinergics which can increase the risk of hyperthermia. Phentermine/topiramate may cause an increased resting heart rate of up to 20 bpm.

Naltrexone-Bupropion

Mechanism of Action



Bupropion is a weak dopamine and norepinephrine reuptake inhibitor, which stimulates pro-opiomelanocortin (POMC) production and release of alpha-MSH and beta-endorphin, that is an endogenous opioid, in vitro. The alpha-MSH activates the melanocortin-4 receptor (MC4R), which acts to reduce food intake, increasing energy expenditure, and causing weight loss. Naltrexone that is an opioid antagonist, works by blocking the mu-opioid receptor and disrupting the beta-endorphin inhibitory feedback on the POMC cells.

Pharmacokinetics

The half-life of naltrexone is 5 hours, and that of bupropion is approximately 21 hours. Naltrexone/bupropion administration is not recommended following a high-fat meal due to a significant increase in systemic exposure.

Adverse Effects

The most common side effects of the naltrexone-bupropion combination are nausea, headache, and constipation. FDA warns to monitor patients for suicidal behavior and ideation, risk of seizure due to bupropion lowering the seizure threshold, increase in blood pressure and heart rate, hepatotoxicity observed with naltrexone exposure, and angle-closure glaucoma.

Orlistat

Mechanism of Action

Orlistat acts by reversibly inhibiting gastric and pancreatic lipases. These lipases have an important role in the digestion of dietary fat. They work by breaking down the triglycerides into absorbable free fatty acids and monoglycerides. Orlistat covalently binds to the serine residues of active sites of lipases and inactivates them. The inactivation of lipases prevents the hydrolysis of triglycerides, and thus free fatty acids are not absorbed.

Pharmacokinetics

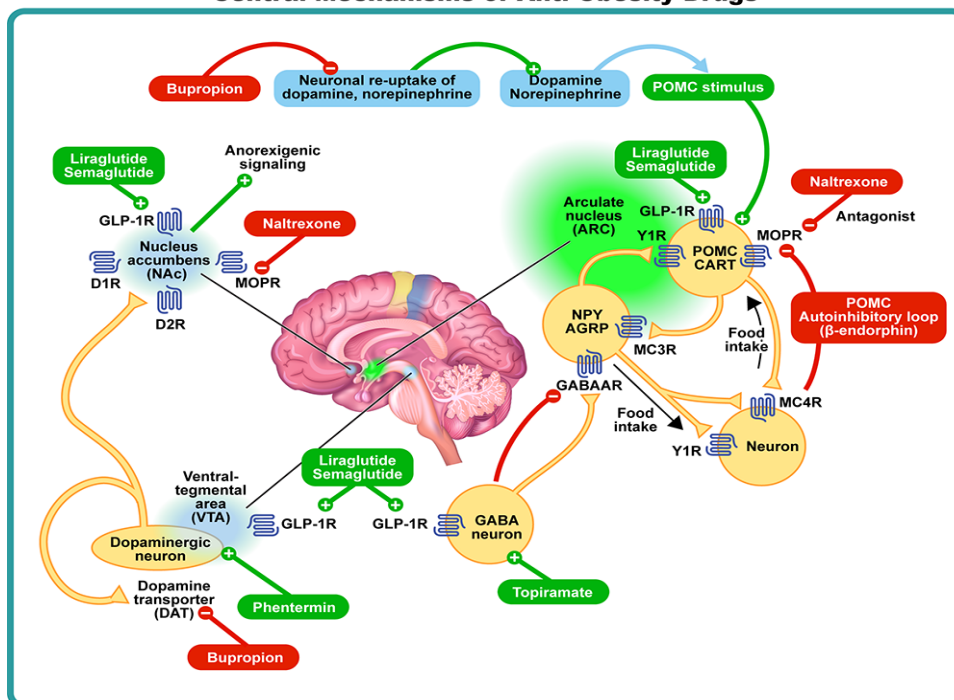
Orlistat acts mainly via its local effect in the gut, and systemic exposure to the medication is minimal. Orlistat metabolism is primarily within the intestinal wall. 95 to 97% of the medication is unabsorbed and excreted in feces.

Adverse Effects

The most common side effect of orlistat use is steatorrhea, which occurs because of the impaired absorption of dietary fat. Other side effects include fecal spotting, diarrhea, abdominal pain, and anal fissures.

Summary: Mechanisms of Action of Anti-Obesity Drugs

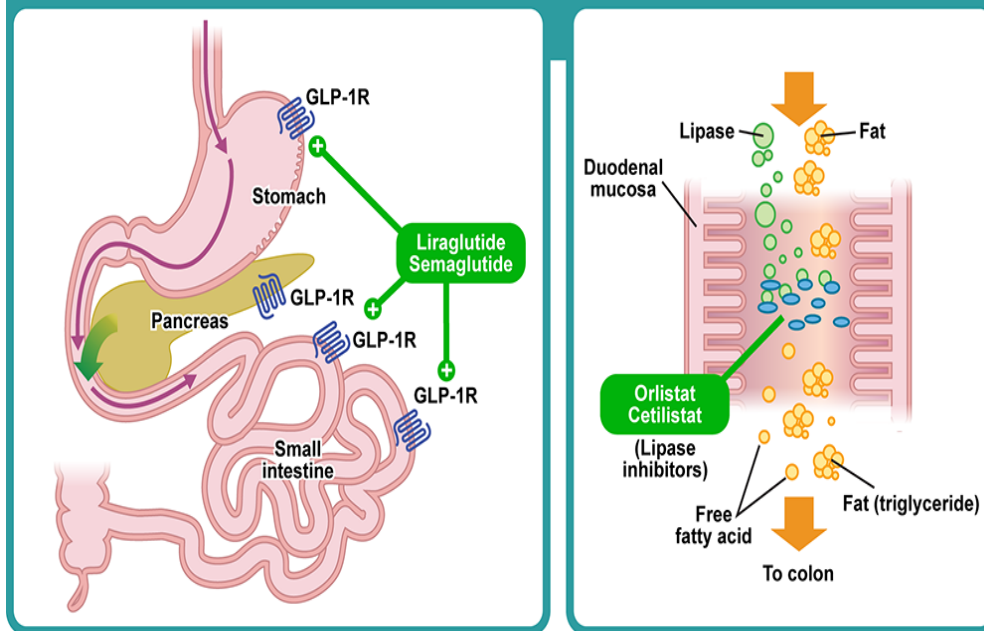
Central Mechanisms of Anti-Obesity Drugs



Central Mechanisms of Anti-Obesity Drugs. Image Credit:

Peripheral Mechanisms of Anti-Obesity Drugs

The peripheral mechanisms of anti-obesity drugs focus on altering metabolic processes, fat absorption, and energy expenditure in the body to support weight loss and improve overall metabolic health.



Peripheral mechanisms of anti-obesity drugs. Image Credit:

Obesity Treatment: Lifestyle, Behavioral, and Medications Part 4 of 4: Clinical Skills

Overview

Learning Objectives

This section covers the following learning objectives:

1. Define metabolic syndrome.
2. Explain the clinical presentations of metabolic syndrome.
3. Explain the pathophysiology of metabolic syndrome.
4. Describe the diagnosis of metabolic syndrome.

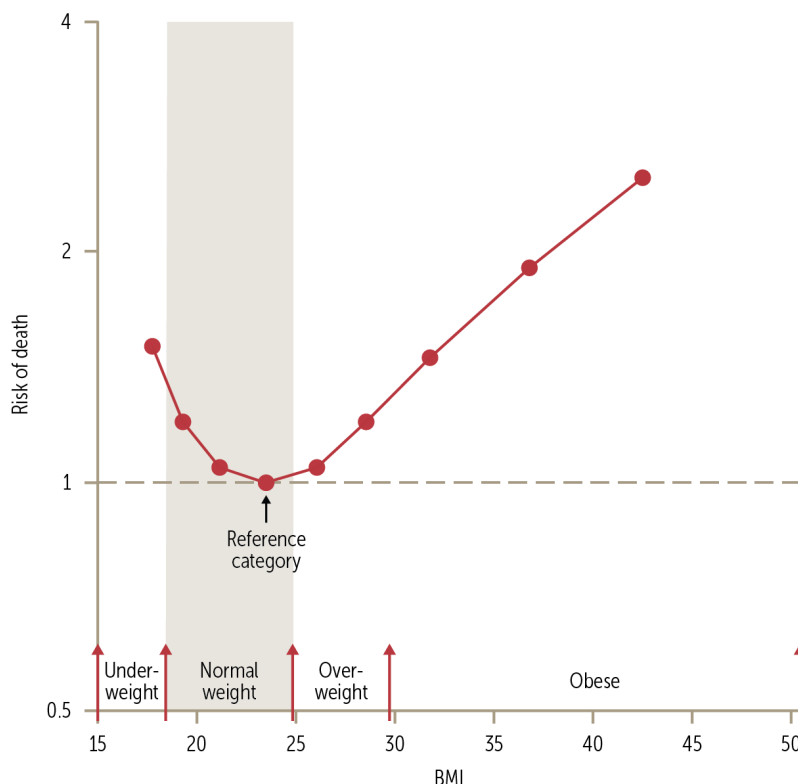
Case Connection

You are seeing JG, a 58-year-old male, 1 day after his emergency department visit for a self-limited episode of abdominal pain. His CT scan did not show any masses nor evidence of bowel obstruction, but the radiologist noted significant fat surrounding the abdominal organs. You tell JG that this is concerning. JG responds, "Well, you don't need a CT to tell you I'm fat. What's the big deal?" On exam, JG's body mass index (BMI) is 32.4 kg/m², his waist circumference is 42 inches, and his blood pressure is 132/88 mm Hg. The remainder of the exam is normal.

What diagnostic testing would you order for JG? What treatment plan would you outline? Consider these questions as you read, and we'll revisit JG at the end of the session.

What Are Obesity and Metabolic Syndrome

Obesity and metabolic syndrome are two very common disorders that account for many hospitalizations and deaths. Being overweight or obese puts patients at risk for type 2 diabetes mellitus (T2DM), hypertension, heart disease, and stroke, among other health problems. The more overweight a person is, the greater is their risk of death, as shown in the image below, in which **body mass index (BMI)** is plotted against the relative risk of death.



BMI and Relative Risk of Death. Image Credit: ScholarRx, used with permission.

Note in the figure above that both underweight and overweight people have increased mortality, but the reasons are different. Underweight people are often so because of a disease that causes increased mortality, such as cancer. On the other hand, evidence suggests that being overweight is itself a cause of increased mortality due to the diseases mentioned above.

Obesity

Obesity is defined as the accumulation of adipose tissue to the point of impairing health. Clinically, it is usually defined by BMI, which is calculated as body weight in kilograms divided by the height in meters squared. Other methods include measuring waist circumference and peripheral and central fat mass, but BMI is the most commonly used classification tool. Note that the BMI screens for weight categories that may lead to health problems, but it does not indicate body fat content or the individual's overall health.

Individuals are allocated to six categories according to BMI:

- Underweight: $<18.5 \text{ kg/m}^2$
- Normal range: $18.5\text{-}24.9 \text{ kg/m}^2$
- Overweight: $25.0\text{-}29.9 \text{ kg/m}^2$
- Class 1 obesity: $30.0\text{-}34.9 \text{ kg/m}^2$



- Class 2 obesity : 35.0-39.9 kg/m²
- Class 3 obesity : >40.0 kg/m²

Because BMI does not give a precise idea of body composition or fat distribution, it might result in a misclassification of individuals with a very high or very low muscle mass percentage. However, BMI is simple to calculate and is now internationally accepted and used as a measure of overall body fat.

Metabolic Syndrome

Obesity, particularly central obesity (i.e., increased abdominal fat, especially around the abdominal viscera), increases the risk for a number of conditions, including T2DM and cardiovascular disease. Obesity is the main driver of a cluster of physiologic abnormalities known as **metabolic syndrome**, which is characterized by visceral or intra-abdominal adiposity, insulin resistance, hyperinsulinemia, glucose intolerance, hypertension, hypertriglyceridemia, and low HDL.

Classifying a patient as having metabolic syndrome has many practical uses, the most important of which is recognizing an increased risk for developing cardiovascular diseases and T2DM.

Many definitions for metabolic syndrome exist, and the criteria to meet them are constantly being revised. According to the American Heart Association/National Heart, Lung, and Blood Institute (AHA/NHLBI), metabolic syndrome is defined as **three or more** of the following criteria:

- Waist circumference >40 inches in males and >35 inches in females
- Systolic blood pressure >130 mm Hg, diastolic blood pressure >80 mm Hg, and/or antihypertensive therapy
- Fasting triglyceride (TG) level >150 mg/dL
- Fasting HDL <40 mg/dL in males and <50 mg/dL in females
- Fasting serum glucose >100 mg/dL

Epidemiology

Obesity is estimated to affect more than a third of the world's population and about 42% of adult population in the United States in 2017 to 2018. US ethnicity-based differences in obesity (i.e., rates are higher in Black and Hispanic people) are likely largely a result of patterns of food intake, lack of exercise, and socioeconomic disadvantage. As an example, both males and females with college degrees tend to have a lower prevalence of obesity.

About one-third of US children are now either obese or overweight. It's more common in children of color than in non-Hispanic White children and worldwide, is more common in wealthy countries. In children, 80% of those with parents with obesity will themselves become obese, likely because of both genetic and socioeconomic factors.

The prevalence of metabolic syndrome is similar to that of obesity and is also related to age, sex assigned at birth, ethnicity, and socioeconomic factors.

Check your knowledge by completing the following exercise.

Knowledge Check Question

Question: What conditions characterize metabolic syndrome?

Answer: Obesity, insulin resistance, high cholesterol, and hypertension characterize metabolic syndrome.

How Do Patients With Obesity or Metabolic Syndrome Present?

A patient with obesity or metabolic syndrome may report undesired weight gain or feeling too heavy, but they are mostly asymptomatic because associated hypertension, high cholesterol levels, and mildly elevated blood sugars do not cause symptoms. A patient with obesity may present with orthopedic problems (e.g., back pain, joint pain) or limited mobility or ambulation. Many will report no specific problems, as they may have lived with their obesity for some time.

In the setting of associated complications like heart disease, patients may have angina or shortness of breath. Patients who have progressed to T2DM might have polydipsia and polyuria, but are more often diagnosed with T2DM by routine lab testing.

Check your knowledge by completing the following exercise.

Knowledge Check Question

Question: How do most patients with obesity or metabolic syndrome present?

Answer: Patients with obesity or metabolic syndrome may report undesired weight gain or feeling too heavy, but they are mostly asymptomatic because the associated hypertension, high cholesterol levels, and mildly elevated blood sugars do not cause symptoms.

What Is the Pathophysiology of Obesity and Metabolic Syndrome

Obesity results from combined genetic and environmental factors. Many people with obesity have an elevated “set point” of normal body weight that makes it hard to achieve sustained weight loss, because the basal metabolic rate will slow to preserve the weight at that set point. This set point may be genetically determined. However, most patients with obesity also have energy intake (calories) that exceeds expenditures, a balance that can be improved by diet and exercise.



Some people can be identified early in life as being at risk for obesity and metabolic syndrome. This includes low-birth-weight infants and children who gain weight rapidly in their first 10 years. Weight should be followed in all patients, because even a small amount of weight gain (<10 pounds) after age 18 increases the risk of heart disease and diabetes mellitus.

Causes

Obesity can be primary or secondary.

Primary Obesity. These patients have a complex mix of genetic and socioeconomic determinants, including:

- Behavioral: overeating, consuming high-fat or high-calorie diets, infant feeding patterns, binge eating, a sedentary lifestyle
- Genetic: a mix of inherited tendencies toward being overweight
- Aging: lower basal metabolic rate, lower activity

In addition, two hormones secreted from adipose tissue are now implicated in some patients with primary obesity. Adiponectin increases energy expenditure, and these hormone levels seem to be reduced in patients with obesity. Leptin acts on the hypothalamus to give a sense of satiety. Some patients with obesity may have leptin resistance, preventing them from regulating their food intake.

Secondary Obesity. A smaller number of patients have obesity because of underlying disease, including:

- Endocrine: polycystic ovary syndrome, hypothyroidism, growth hormone deficiency, Cushing syndrome (high cortisol)
- Medications: insulin (in T2DM), antipsychotics (eg, thioridazine, clozapine, olanzapine), antidepressants (tricyclics), and antiseizure medications (valproate, carbamazepine, gabapentin)
- Rare congenital syndromes such as Prader-Willi and Bardet-Biedl

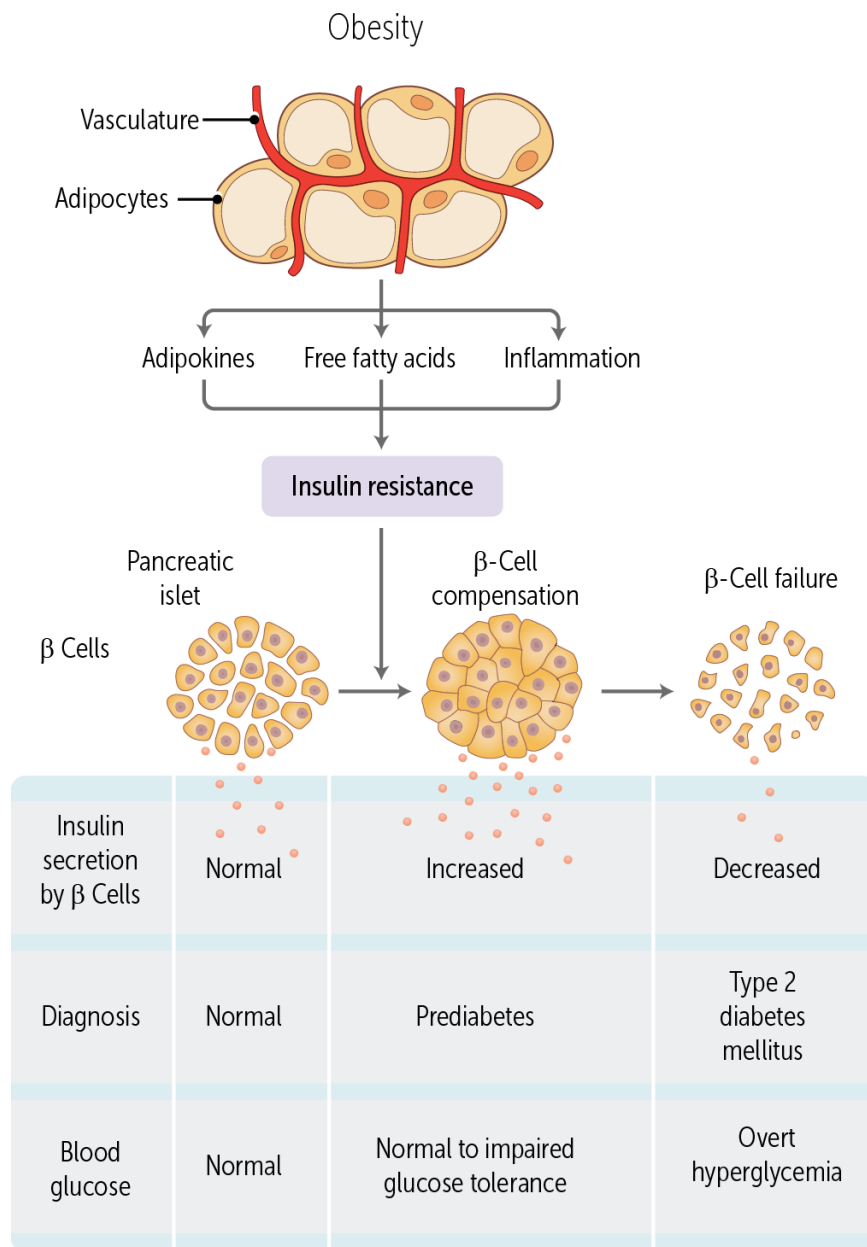
Complications

Metabolic complications of obesity and metabolic syndrome include insulin resistance, T2DM, and ischemic heart disease (caused by hyperlipidemia and atherogenesis). Most of these complications, especially insulin resistance, are more frequent in those with **central (abdominal) obesity**. In fact, patients with increased waist size are at risk for these complications even if their overall BMI is normal. Visceral fat seems to carry a greater risk than does subcutaneous fat in the buttocks and thighs. The reason for this increased risk is not known; however, each different type of fat produces different adipokines.

Insulin Resistance. Insulin is produced by the pancreas in response to hyperglycemia and stimulates glucose use in tissues such as muscle, fat, and liver. Normally, insulin translocates the glucose transporter type 4 (GLUT4) from the cytosol to the cell surface and induces glucose uptake.

Insulin resistance occurs when tissues have decreased responsiveness to the effects of insulin. This leads to a reduction in the tissues' glucose uptake, which increases the glucose levels in the

blood (hyperglycemia) and can eventually lead to T2DM. The precise cause of insulin resistance is still unclear; however, obesity (particularly abdominal obesity and increased waist circumference) is associated with insulin resistance. Increased inflammation, including secretion of adipokines, results (see figure below).



Obesity and Insulin Resistance. Image Credit: ScholarRx, used with permission.

Insulin resistance leads to pancreatic β -cell hypertrophy with mildly elevated blood glucose levels and increased insulin secretion. This “prediabetes” is typical of metabolic syndrome, in which glucose levels are elevated to subdiabetic levels. As shown in the figure above, eventually serum glucose levels may rise into the diabetic range, after which β -cell failure occurs, and the patient develops overt T2DM.



Check your knowledge by completing the following exercise.

Knowledge Check Question

Question: Which three tissues respond to insulin and use glucose?

Answer: The three tissues that respond to insulin and use glucose are muscle, adipose, and liver tissues.

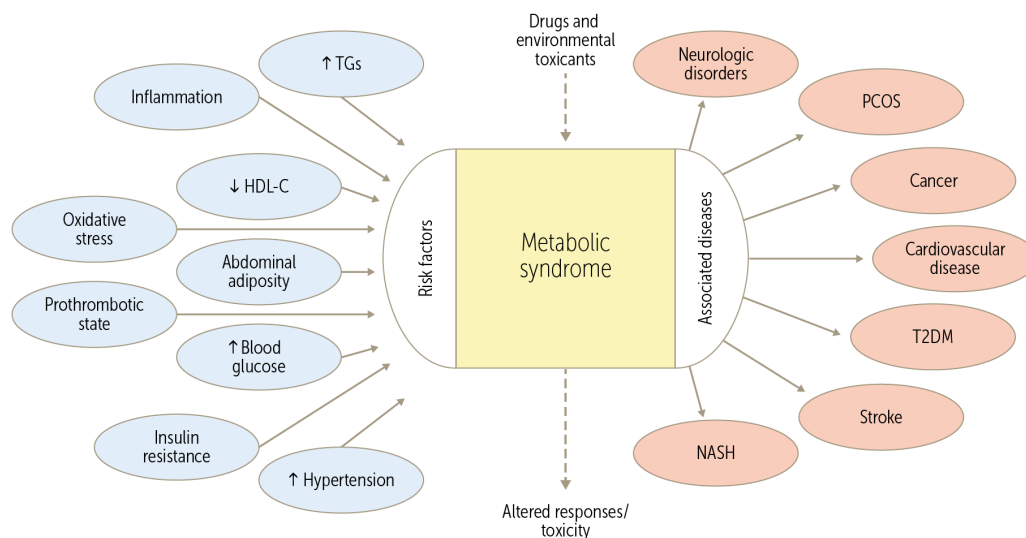
Hyperlipidemia and Atherogenesis. Insulin resistance leads to high serum lipids (dyslipidemia) because insulin typically suppresses lipolysis in adipocytes. Therefore, decreased insulin signaling leads to increased lipolysis, which results in higher free fatty acid (FFA) levels.

This abnormal lipid profile leads to **atherogenic dyslipidemia**, characterized by high levels of plasma TGs, very-low-density lipoprotein (VLDL), and LDL, as well as low levels of HDL. Specifically, insulin resistance increases the production of VLDL and decreases its clearance because of reduced levels of the enzyme lipoprotein lipase (LPL). LPL synthesis is regulated by insulin and is reduced in insulin resistance because of decreased insulin signaling.

These events together result in endothelial dysfunction, which plays a major role in the development of atheromas. Atheroma formation in the blood vessels narrows them and leads to complications such as coronary artery disease, stroke, and peripheral artery disease.

Other Metabolic Complications. Other complications of the metabolic syndrome include (see figure below):

- Cardiovascular disease (stroke, heart failure, ischemic heart disease)
- Neurologic disorders like diabetic peripheral neuropathy (this can occur in metabolic syndrome even if patients have not yet progressed to overt diabetes mellitus)
- Reproductive/metabolic disorders like polycystic ovary syndrome (PCOS)
- Many cancers (eg, endometrial, pancreatic)
- Liver disease (nonalcoholic steatohepatitis [NASH]; “fatty liver”)



Complications of the metabolic syndrome. Image Credit: ScholarRx, used with permission.

Some of these complications have known pathophysiologic mechanisms that explain the link to obesity and metabolic syndrome. These include increased inflammation and oxidative stress on cells, leading to their damage (e.g., endothelial cells, neurons). However, for some complications, such as PCOS and cancer, the pathophysiologic mechanism has not been clearly identified.

Role of the Gut Microbiome

Recent research has shown that the gut microbiome (gut flora) has an impact on nutrient metabolism, energy expenditure, and the appetite center in the brain. Some drugs that are effective in treating obesity do so by changing the composition of the gut microbiome. For example, supplementation with probiotics (live bacteria and yeast), prebiotics (plant fibers that act as food for the gut flora), and synbiotics (mixture of pro- and prebiotics) may alter the secretion of hormones, neurotransmitters, and inflammatory factors, thus preventing food intake triggers that lead to weight gain. How the different species of bacteria in the gut microbiome precisely impact weight gain and how using the supplements can lead to weight loss are not yet clear.

Other Consequences of Obesity

Some of the main reasons that patients who are overweight or obese present to their physicians are for orthopedic, respiratory, biliary, gastrointestinal (GI), urinary, and reproductive problems.

Orthopedic Problems. Both osteoarthritis and gout occur more often in patients with obesity. In addition, carrying more weight puts increased stress on bones and joints, so it



can worsen pain in many joint and bone diseases. In osteoarthritis, for example, losing weight can significantly lower pain scores in patients with knee or hip disease.

Children who are obese may develop a slipped capital femoral epiphysis in which the epiphysis is displaced from the femoral plate. They can also develop bowed legs (genu varus).

Respiratory Problems. Obstructive sleep apnea (OSA) and obesity hyperventilation syndromes are both associated with obesity, causing daytime sleepiness, snoring, and choking during sleep. OSA is an independent risk factor for coronary artery disease.

Biliary and GI Tract Problems. Obesity increases the incidence of gallstones and the rates of gastroesophageal reflux disease and esophageal and gastric cancers.

Urinary Problems. Obesity increases the incidence of incontinence and, probably, kidney stones. Severe obesity is associated with a form of glomerular disease (focal segmental glomerulosclerosis) resulting from high intrarenal pressures.

Reproductive/Sexual Problems. Obesity may cause irregular menses and increase the rate of complications during pregnancy. For example, females with obesity are more likely to require cesarean deliveries. In males, obesity is a risk factor for erectile dysfunction.

How Do We Diagnose Obesity and Metabolic Syndrome

As mentioned, the most widely used way to diagnose obesity is by BMI. Other methods are waist circumference, waist-to-hip ratio, waist-to-height ratio, and skinfold measurement. The waist circumference approximates intra-abdominal fat. The gold standard for diagnosis (not used often because of cost) is the measurement of visceral fat at the L4/L5 level by abdominal CT or magnetic resonance imaging.

Check your knowledge by completing the following exercise.

Knowledge Check Question

Question: What are common clinical tools used to diagnose obesity?

Answer: Five common methods to diagnose obesity are BMI calculation, waist circumference measurement, waist-to-hip ratio, waist-to-height ratio, and skinfold measurement.

Metabolic Syndrome

Specific laboratory tests and clinical examinations are used to diagnose metabolic syndrome. These include blood pressure measurement and blood tests for **TGs, fasting plasma glucose, and cholesterol levels**. The AHA/NHLBI require at least three of the following criteria to diagnose metabolic syndrome:



- Abdominal obesity measured by the waist circumference (>40 inches in males and >35 inches in females)
- Hypertension (systolic blood pressure >130 mm Hg, diastolic blood pressure >80 mm Hg) or on medications for blood pressure
- TGs $\geq$ 150 mg/dL or on medications for hypertriglyceridemia
- Low HDL cholesterol: <40 mg/dL in males and <50 mg/dL in females or on medications for low HDL
- Fasting plasma glucose $\geq$ 100 mg/dL or history of T2DM or on medications for hyperglycemia

How Do We Manage Obesity and Metabolic Syndrome?

Obesity and metabolic syndrome are serious medical problems that should be addressed with all patients at risk. All patients who have a BMI of $>25 \text{ kg/m}^2$ should be advised about lifestyle modifications, including diet and exercise. Medical therapy and surgery are then used if lifestyle modifications are insufficient.

Lifestyle Modifications

Diet and increased activity are the keys to weight loss, although diet is more important.

Diet. Reducing caloric intake is key to weight loss. The normal recommended caloric intake for nonobese people ranges from 1600 to 2400 kcal/day in females (depending on age and activity) and about 2000 to 3000 kcal/day for males. However, for patients with obesity, the recommended intake is reduced to 800 to 1200 kcal/day. This is often hard to maintain. Another approach is to reduce daily intake by 500 kcal/day, which is more sustainable for many patients.

Some diets, such as the Mediterranean diet and a low-fat, low-carbohydrate diet can also work without such marked caloric restriction. Other diets, such as the ketogenic diet and intermittent fasting, should be explored under clinical observation.

Clinical Correlation

The Mediterranean diet is high in vegetables, whole grains, fruits, beans, nuts, and seeds and uses olive oil as the main source of fat.

Increased Activity. Although it is less effective overall than diet modification, exercising for 30 minutes or more 5 to 7 days per week, is effective in promoting weight loss.

Drug Therapy



Medications can be directed at weight loss and at avoiding the complications of metabolic syndrome.

Weight Loss. Some patients benefit from drugs that enhance weight loss. This includes patients who have a BMI greater than 30 kg/m² and have not been able to lose weight as well as those with severe complications of obesity. Some drug options include:

- Glucagon-like peptide 1 (GLP-1) agonists such as semaglutide. These drugs increase pancreatic insulin secretion and promote weight loss regardless of whether the patient has overt T2DM. The weight loss likely occurs both because of appetite suppression and adverse drug effects such as mild nausea.
- Orlistat inhibits pancreatic lipase, so it blocks the absorption of dietary fats.
- Phentermine is used for short-term weight reduction; this sympathomimetic agent (similar to amphetamines) raises catecholamine levels and body energy expenditure.
- Hydrogels are ingested gels that expand in the stomach to promote a feeling of fullness.

Avoiding Obesity Complications. Pharmacologic interventions can improve the levels of cholesterol, TGs, glucose, and blood pressure that are abnormal in metabolic syndrome. Metformin and glitazones have been shown to reduce insulin resistance and improve glucose levels. A combination of statins, fibrates, and niacin is commonly recommended according to the patient's lipidemic profile and their overall needs:

- Statins lower LDL levels; in higher doses, they lower TGs and elevate HDL levels.
- Fibrates lower LDL levels and raise HDL.
- Niacin is the drug of choice for raising low HDL.

A variety of different medications, including angiotensin-converting enzyme inhibitors and angiotensin receptor blockers, are used to lower blood pressure and improve hypertension.

Devices and Surgery

For patients who fail the above treatments and have very severe obesity, bariatric surgery is possible. The types include:

- Gastric banding, in which a band is placed laparoscopically around the upper part of the stomach to reduce its size
Intragastric balloons.
- Electrical stimulation to block vagus nerve signals to the stomach.
- Shortening the intestinal surface area as in small intestine bypass and/or restriction (by reducing the stomach size as in sleeve gastrectomy)



Case Connection

Thinking back to JG, what diagnostic testing would you order? What treatment would you outline?

JG has class 1 obesity (BMI 30.0-34.9 kg/m²). Obesity, particularly central obesity, puts him at increased risk for T2DM and cardiovascular disease. A fasting lipid profile and fasting glucose reveal total cholesterol of 252 mg/dL (normal <200 mg/dL), TGs of 222 mg/dL (normal <150 mg/dL), HDL of 32 mg/dL (normal >40 mg/dL), and glucose of 122 mg/dL (normal <100 mg/dL). You see that JG has all five criteria for metabolic syndrome: waist circumference >40 inches, blood pressure >130/85 mm Hg, fasting TGs >150 mg/dL, HDL <40 mg/dL, and fasting glucose >100 mg/dL. After explaining the significance of metabolic syndrome to JG, you discuss lifestyle modifications (diet and increased activity) as the first line of treatment. You also discuss the potential future role of medications and bariatric surgery. "You know me. I don't like pills, and I certainly don't want surgery," says JG. "Let's try this diet thing first and see how it goes." At his 6-week follow-up, JG has lost 10 pounds. "I'm willing to keep going," he says. "Dieting is really not as hard as I thought, but I do miss potato chips."

Weight Control & Bariatric Surgery Professionalism- Check your Bias!

Weight Bias

"Weight bias is commonly observed by students in health disciplines, who themselves report frustrations and stereotypes about treating patients with obesity" (Puhl et al, 2014 p.1).



Patient with Morbid Obesity. Image Credit: Tanaka, T., Kita, Y., Mori, S. et al. Three-stage laparoscopic surgery in a morbidly obese patient with Hinchey III diverticulitis: a case report. *surg case rep* 5, 24 (2019). <https://doi.org/10.1186/s40792-019-0588-7> CC-BY-4.0

Impact on Providers

- Less patient centered care with implicit bias against stigmatized groups
- View patients as “lazy, weak willed, non-adherent”
- Allocate less time with patients with obesity vs. non obese
- Overlook other diagnosis and evaluations
- Patients with obesity more likely to get lifestyle counselling vs. meds (MS study with virtual patients)

Impact on Patients

- Higher levels of stress (allostatic load) which over time can lead to heart disease, stroke, depression, anxiety
- Unhealthy eating behaviors, binge eating disorder, and lower motivation for exercise
- Reduced engagement and trust of healthcare system
 - Less likely to get routine screenings (mammogram)
 - Less likely to remember instructions

Case Connection

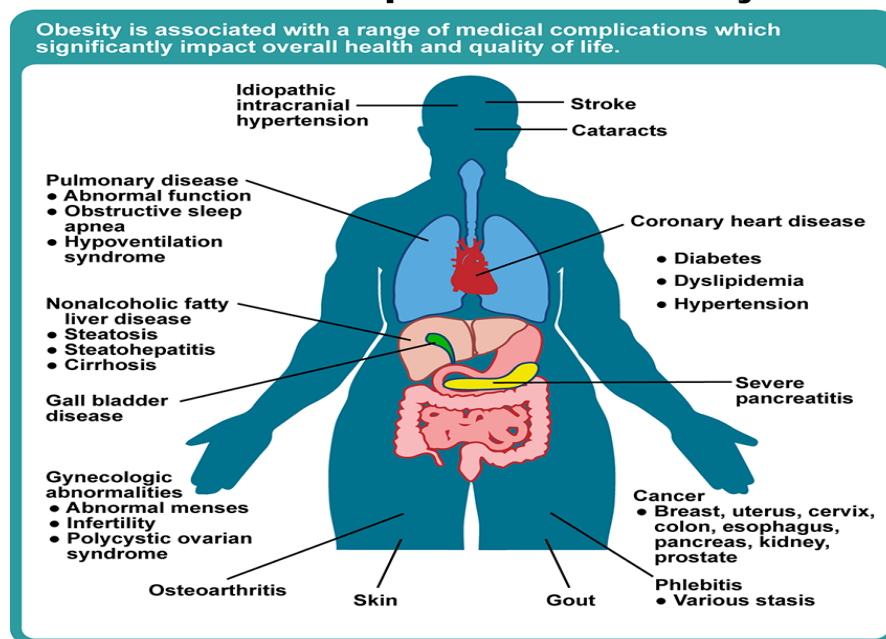
A 39-year-old man presents to his family care physician for his suitability for 'weight loss' surgery. He describes a 10-year history of fluctuation of his body weight. He is concerned about his family history of hypertension, especially since he became a father 1 month ago. He has tried several weight-reduction programs but has not met his target weight. He denies any history of sleep apnea, diabetes mellitus or hypercholesterolemia.

On examination, he is alert and undistressed. His body mass index (BMI) is calculated to be 41 kg/m^2 . His vital signs are pulse 88/min, blood pressure 110/70 mmHg, temperature 98.6°F (37°C), respirations 18/min. Cardiovascular exam and respiratory exam are unremarkable.

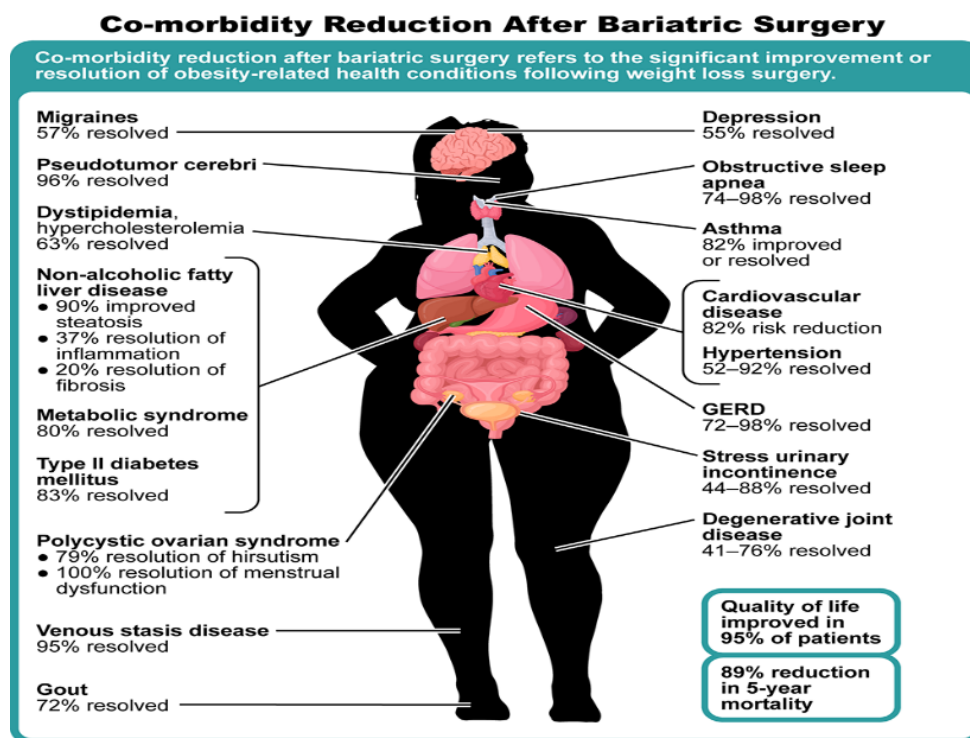
Which of the following is most likely to make the patient qualify for bariatric surgery?

Why Is Bariatric Surgery Performed?

Medical Complications of Obesity



Medical Complications of Obesity. Image Credit: Adtale



Co-Morbidity Reduction After Bariatric Surgery. Image Credit: Aadtalem

Bariatric surgery helps the patient to lose excess weight and thus reduce the risk of potentially life-threatening weight-related health problems; such as: GERD, CVA, CAD, DM, Sleep Apnea

- Metabolic and hormonal benefits independent of weight loss
- Weight loss is greater and more sustained than nonsurgical intervention alone.
- Ideally, the patient should lose 33% to 50% of weight within 2 years after surgery. Studies show ~25% weight loss 10-12 years post op.
- Weight-loss surgeries are typically done only after failure of other weight reduction procedures.

Eligibility Criteria for Bariatric Surgery

Eligibility criteria (NIH 1991) for surgery are primarily based on body mass index (BMI) and the presence of obesity-related complications, including:

- individuals with BMI $>40 \text{ kg/m}^2$
- individuals with BMI $>35 \text{ kg/m}^2$ with one or more obesity-related complications, including type 2 diabetes, hypertension, hyperlipidemia, and obstructive sleep apnea

Recently (2022), several organizations (ASMBS) recommend bariatric surgery for

- all individuals with BMI $\geq 35 \text{ kg/m}^2$
- individuals with T2D and BMI $> 30 \text{ kg/m}^2$



Calculating BMI

$$BMI = \frac{weight(kg)}{height^2(m^2)}$$

$$BMI = \frac{weight(lb) \cdot 703}{height^2(in^2)}$$

$$BMI = \frac{weight(lb) \cdot 4.88}{height^2(ft^2)}$$

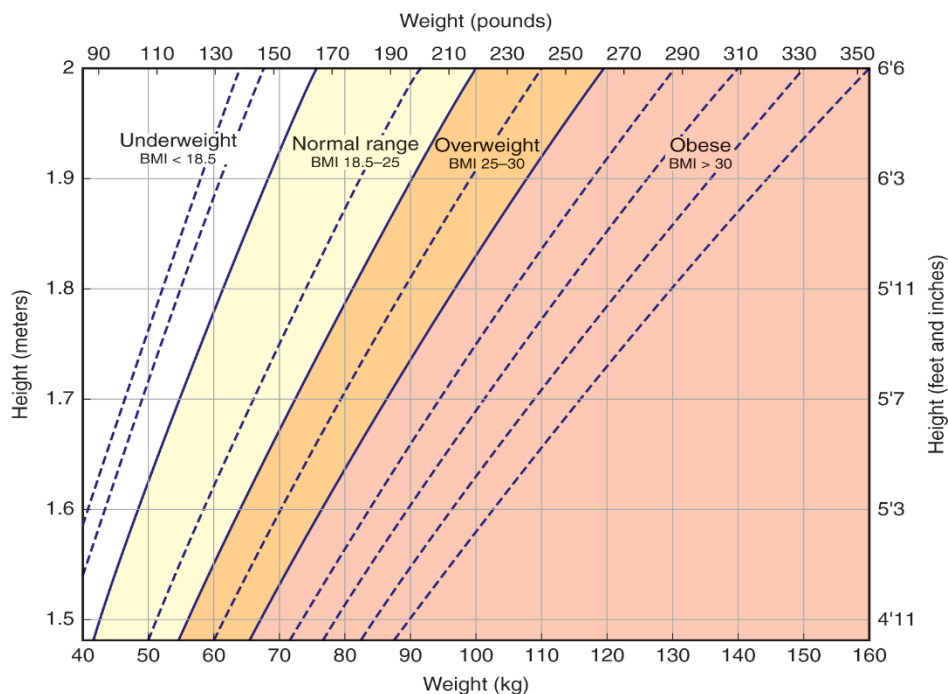


Chart for estimating body mass index (BMI). To find the BMI category for a particular subject, locate the point at which the height and weight intersect. Image credit: McGraw Hill, used with permission.

Classification	BMI (kg/m ²)	Risk of Comorbidities
Underweight	< 18.5	Low (but risk of other clinical problems increased)
Normal range	18.5 to 24.9	Average
Overweight	≥ 25	
Pre-Obese	25.0 to 29.9	Increased



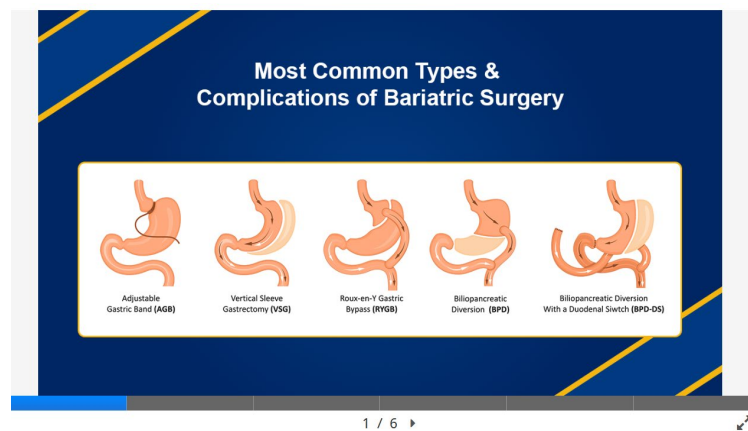
Classification	BMI (kg/m ²)	Risk of Comorbidities
Obese Class 1	30.0 to 34.9	Moderate
Obese Class 2	35.0 to 39.9	Severe
Obese Class 3	≥ 40.0	Very Severe
Waist Circumference (cm)		
Comorbidity Risk	Women	Men
Above action level 1	≥80	≥94
Above action level 2	≥88	≥102

Steps Before the Surgery

- Extensive screening process- consultations and diagnostic testing.
- Patient must be willing to make permanent changes that lead to a healthier living.
- Patient must accept a long-term follow-up plan that includes monitoring nutrition, lifestyle, and medical conditions.
- Patient should be aware of that they may need additional surgery for reductant skin/tissue
- Bariatric surgery is Not body contouring surgery
- Patient should be informed that these are all expensive surgical procedures for which complications exist.

Most Common Types of Bariatric Surgery

Refer to the Canvas page for the full interactive experience.



Case Connection

A 39-year-old man presents to his family care physician for his suitability for 'weight loss' surgery. He describes a 10-year history of fluctuation of his body weight. He is concerned about his family history of hypertension, especially since he became a father 1 month ago. He has tried several weight-reduction programs but has not met his target weight. He denies any history of sleep apnea, diabetes mellitus or hypercholesterolemia.

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Which of the following is most likely to make the patient qualify for bariatric surgery?

- The patient's family history of hypertension
- The patient's blood pressure reading
- The patient's BMI
- The patient's gender
- The patient recently becoming a father



Summary

- Obesity is defined as a body mass index (BMI) of ≥ 30 kg/m².
- Metabolic syndrome is the co-occurrence of insulin resistance, obesity, atherogenic dyslipidemia, and hypertension.
- Patients with obesity or metabolic syndrome present with weight gain but are mostly asymptomatic.
- Xanthelasma is seen in severe dyslipidemia.
- Obesity and the metabolic syndrome are the result of a complex mix of genetic and behavioral factors.
- Obesity and metabolic syndrome may have orthopedic, respiratory, biliary, gastrointestinal, urinary, and reproductive implications.
- Obesity and metabolic syndrome are associated with insulin resistance, which leads to increased inflammation.
- The diagnosis of obesity is based on BMI.
- The diagnosis of metabolic syndrome is based on fasting glucose levels, hypertension, TGs, cholesterol, and waist circumference.
- Lifestyle changes should be recommended, including diet and exercise.
- Some patients benefit from drugs such as semaglutide, orlistat, and phentermine.
- For patients who fail treatments and have very severe obesity (BMI >40 kg/m²), surgery and implantable device options exist.

Check Your Knowledge

Obesity Treatment: Lifestyle, Behavioral, and Medication Part 4: Clinical Skills - Review Questions

Question 1

What is the body mass index threshold for obesity (in kg/m²)?

- 25
- 30
- 30 for females and 32 for males
- 35

Question 2

Which of the following is a common complication of metabolic syndrome?

- Alzheimer disease
- Coronary artery disease
- Multiple sclerosis
- Rheumatoid arthritis
- Schizophrenia



Question 3

Which of the following serum studies are included in the diagnosis of metabolic syndrome?

- HDL, LDL, and apolipoprotein B
- TGs, cholesterol, and fasting plasma glucose
- TGs, HDL, and prothrombin time
- Transaminase, cholesterol, and anti-GLUT4

Question 4

Which of the following treatment for obesity is most appropriate for a sedentary 80-year-old male with a body mass index of 32 and severe osteoarthritis that limits his mobility

- Gastric banding
- Mediterranean diet
- Semaglutide
- Sleeve gastrectomy
- Treadmill exercise 5-7 days/week



Obesity Treatment: Lifestyle, Behavioral, and Medication Part 4: Clinical Skills - Review Questions and Answers

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- **30 (Correct answer)**
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